

## Unit 1, Lesson 8: Line Segments



### Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.1 Construct a copy of a segment or an angle.

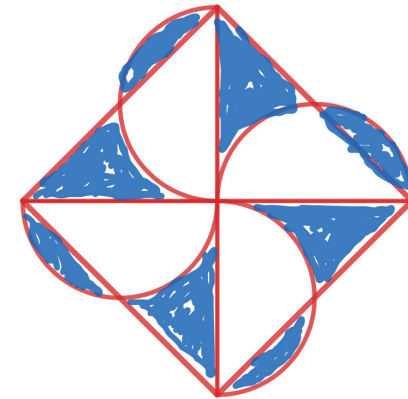


### Learning Targets

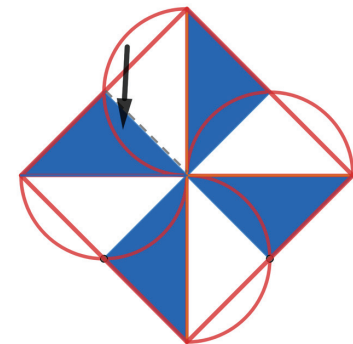
- ☐ I can construct a copy of a line segment.
- ☐ I can use the segment addition postulate.



### 1.8.1 Warm-Up: Semicircles and Triangles

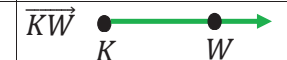


1. What do you notice about the picture?
2. If the radius of each semicircle is 5, then what is the length of each of the square's diagonals?
3. Dina determined the area of the shaded region using the drawing shown. Find the total shaded area.





## 1.8.2 Guided Instruction: Line Segments

| Line                                                                                                                                                                 | Line Segment                                                                      | Ray                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A line is an undefined term.                                                                                                                                         | A part of a line that is bounded by two distinct end points.                      | A part of a line that has a fixed starting point and extends infinitely in one direction.                                                                              |
| <br> |  | <br> |

When naming a line segment, the line segment symbol should be used; however, when giving the length of a line segment, the symbol should not be used.

$$\overline{TM} \quad TM = 12 \text{ millimeters}$$

1.  $\overline{RK}$  is shown. Use a ruler to measure the line segment in centimeters.

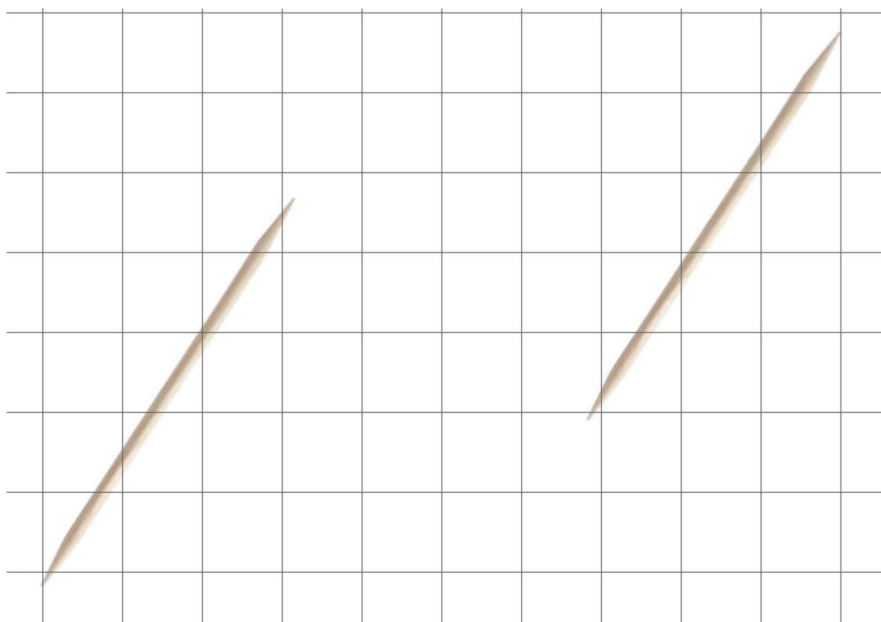


2. Use patty paper to copy  $\overline{LD}$ . Then, draw line segment  $\overline{XN}$  with the same length as  $\overline{LD}$ .



3. Follow these steps to draw a copy of  $\overline{LD}$  using a compass.
  - Draw point  $C$ . This will be one endpoint of the new line segment.
  - Set the compass' point on point  $L$ .
  - Adjust the compass' width to point  $D$  so the compass' width is equal to the length of  $\overline{LD}$ .
  - Without changing the compass' width, place the compass' point on  $C$  and draw an arc roughly where the other endpoint will be.
  - Pick a point  $M$  on the arc that will be the other endpoint of the new line segment.
  - Draw line segment  $\overline{CM}$  using a straightedge.

4. Discuss with your partner how to show that the two toothpicks are congruent using each method in the table. Then, write a brief explanation for each method.



|                       |  |
|-----------------------|--|
| <b>Transformation</b> |  |
| <b>Patty Paper</b>    |  |
| <b>Compass</b>        |  |



### Ruler Postulate

The points of a line can be placed in a one-to-one correspondence with the real numbers such that the distance between two distinct points is the absolute value of the difference of the corresponding real numbers.

5.  $\overline{KC}$  and  $\overline{YC}$  are shown.



- Discuss with your partner how aligning a ruler with  $\overline{KC}$  illustrates the ruler postulate. Summarize your discussion.
- Use a ruler to measure each line segment. Include units.  
  
 $KY = \underline{\hspace{2cm}}$        $YC = \underline{\hspace{2cm}}$        $KC = \underline{\hspace{2cm}}$
- Discuss with your partner why  $KY + YC = KC$  should be true.

Points on the same line are called **collinear points**.



### Segment Addition Postulate

If points  $A$ ,  $X$ , and  $B$  are collinear points and point  $X$  is between points  $A$  and  $B$ , then  $AX + XB = AB$ .

6.  $\overline{KW}$  and  $\overline{WT}$  are shown, where  $KW = 4x - 7$ ,  $WT = 23.5$ , and  $KT = 7x - 3$ .



- a. Determine the value of  $x$ .
- b. What is the length of  $\overline{KW}$ ?



### 1.8.3 Guided Practice: Segment Addition

1.  $\overline{YB}$ ,  $\overline{BW}$ , and  $\overline{WH}$  are shown, where  $YB = 2x$ ,  $BW = x + 4$ ,  $WH = 3x - 1$ , and  $BH = 35$ .



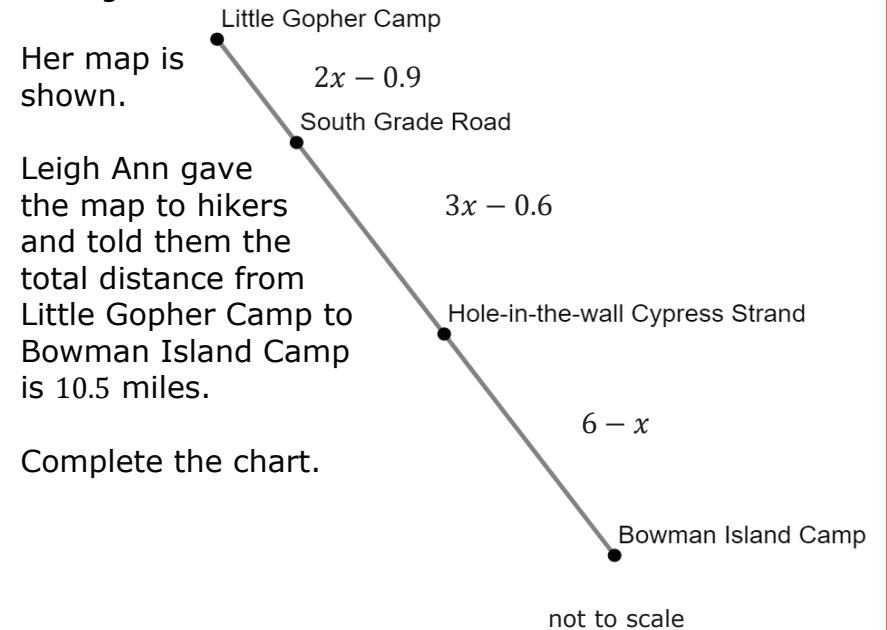
- a. Determine the value of  $x$ .
- b. What is the length of  $\overline{YW}$ ?

2. Points  $K$ ,  $L$ ,  $M$ , and  $N$  are collinear such that  $KM = LN$ .
- Draw a diagram that represents points  $K$ ,  $L$ ,  $M$ , and  $N$ .
  - Compare your diagram with your partner. Discuss whether or not the diagrams drawn must all be the same. Summarize your discussion.
  - Determine the value of  $x$  if  $KM = 6x - 6.2$  and  $LN = 5x - 2$ .
  - Using the information from c, determine the length of  $\overline{KM}$ .



### 1.8.4 Try It: Ocean to Lake Florida Trail

1. The Ocean to Lake Hiking Trail runs for approximately 63 miles, west from Hobe Sound Beach to Lake Okeechobee. Leigh Ann created a simple map of part of the trail that goes through J.W. Corbett Wildlife Management Area.



| Hike                                                  | Distance (miles) |
|-------------------------------------------------------|------------------|
| Little Gopher Camp to South Grade Road                |                  |
| South Grade Road to Hole-in-the-wall Cypress Strand   |                  |
| Hole-in-the-wall Cypress Strand to Bowman Island Camp |                  |



### 1.8.5 Wrap-Up: Cool Down

1. Create an example of the Segment Addition Postulate and show the steps to finding the total length.

## Unit 1, Lesson 9: Copying an Angle



### Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.1 Construct a copy of a segment or an angle.



### Learning Targets

- ☐ I can construct a copy of an angle.
- ☐ I can use the angle addition postulate.